

数值分析第十一次作业

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1. 基于一阶导数的中心差商近似公式

$$f'(x) \approx \frac{1}{2h}(f(x+h) - f(x-h))$$

构造一个对于二阶导数 $\frac{\partial^2 u(x,y)}{\partial x \partial y}$ 的近似, 并分析这一逼近的误差阶

解答: 近似公式为

$$\frac{\partial^2 u(x,y)}{\partial x \partial y} \approx \frac{1}{4h^2}(u(x+h, y+h) + u(x-h, y-h) - u(x-h, y+h) - u(x+h, y-h)),$$

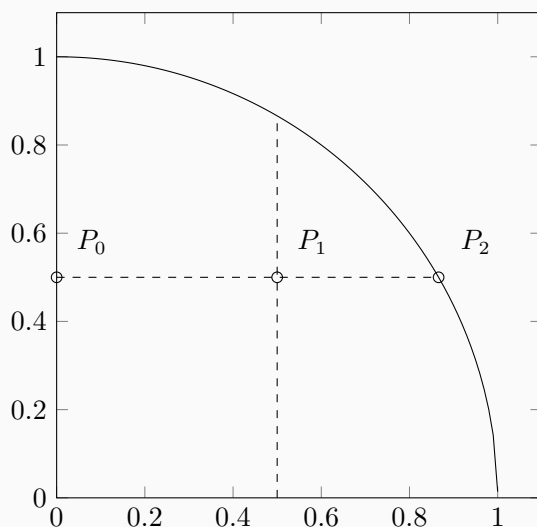
由 Taylor 展开可得逼近的误差为

$$\begin{aligned} & \frac{1}{4h^2}(u(x+h, y+h) + u(x-h, y-h) - u(x-h, y+h) - u(x+h, y-h)) - \frac{\partial^2 u(x,y)}{\partial x \partial y} \\ &= \frac{1}{6} \frac{\partial^4 u(x,y)}{\partial x^3 \partial y} h^2 + \frac{1}{6} \frac{\partial^4 u(x,y)}{\partial x \partial y^3} h^2 + \mathcal{O}(h^4) = \mathcal{O}(h^2), \end{aligned}$$

故误差阶为 2 阶.

□

2. 针对图 1(单位圆在第一象限的部分) 所示, 基于函数 $u(x,y)$ 在 $u_0 = u(P_0), u_1 = u(P_1), u_2 = u(P_2)$ 的值近似 $\frac{\partial u}{\partial x} \Big|_{P_1}$, 并给出误差估计.

图 1: P_0, P_1, P_2 在第一象限处的位置

解答: 设 $x_0 = x_1 - h_1, x_2 = x_1 + h_2$, 由 Taylor 展开可得

$$u_0 = u_1 - h_1 \frac{\partial u}{\partial x} \Big|_{P_1} + \frac{1}{2} h_1^2 \frac{\partial^2 u}{\partial x^2} \Big|_{P_1} + \mathcal{O}(h_1^3), \quad u_2 = u_1 + h_2 \frac{\partial u}{\partial x} \Big|_{P_1} + \frac{1}{2} h_2^2 \frac{\partial^2 u}{\partial x^2} \Big|_{P_1} + \mathcal{O}(h_2^3),$$

消去 $\frac{\partial^2 u}{\partial x^2} \Big|_{P_1}$ 可得

$$\frac{\partial u}{\partial x} \Big|_{P_1} = \frac{1}{h_1 h_2 (h_1 + h_2)} ((h_2^2 - h_1^2) u_1 + h_1^2 u_2 - h_2^2 u_0) + \mathcal{O}(h_1^2 + h_2^2),$$

故近似公式为

$$\frac{\partial u}{\partial x} \Big|_{P_1} \approx \frac{1}{h_1 h_2 (h_1 + h_2)} ((h_2^2 - h_1^2) u_1 + h_1^2 u_2 - h_2^2 u_0),$$

误差为 $\mathcal{O}(h_1^2 + h_2^2)$.

□